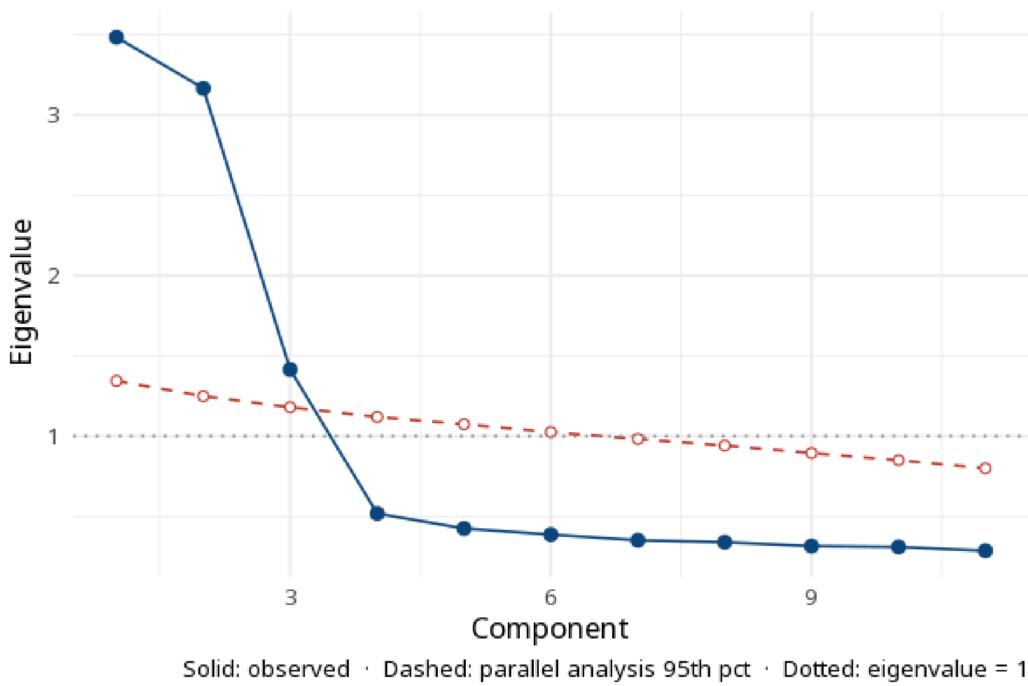
1 Principal component analysis (PCA)

Component	Eigenvalue	% variance	Cumulative %
PC1	3.484	31.7	31.7
PC2	3.167	28.8	60.5
PC3	1.415	12.9	73.3

Variable	PC1	PC2	PC3
esg1	−.54	−.61	
esg2	−.48	−.60	
esg3	−.50	−.60	
esg4	−.47	−.59	
norm1	−.50	.68	
norm2	−.50	.64	
norm3	−.51	.64	
norm4	−.51	.65	
intent1	−.71		−.52
intent2	−.67		−.55
intent3	−.73		−.48



PCA compresses correlated variables into a few components. The variance explained (Table 1) shows how much information each component retains; the component loadings (Table 2) show how strongly each variable contributes.

Loadings are correlation-scaled (eigenvector $\times$ $\sqrt{\text{eigenvalue}}$) so they share the unit of a correlation coefficient. PCA is data reduction — components are weighted composites of the observed variables, not reflective latent factors; do not interpret them as constructs and do not report communalities (which belong to factor analysis, not PCA). For a latent-construct model use EFA (psych::fa) or CFA (lavaan). The scree plot overlays the parallel-analysis 95th-percentile threshold — retain components where the observed eigenvalue exceeds that line; the Kaiser eigenvalue $>$ 1 rule is shown for reference but tends to over-extract (Zwick & Velicer, 1986). Cumulative variance $\geq$ 70% is a common but explicitly subjective target (Jolliffe & Cadima, 2016).

PCA (correlation matrix; parallel analysis) retained 3 components explaining 73.3% of total variance.

Statistical basis: Principal component analysis as a data-reduction method (not a latent-variable model) (Jolliffe, I. T., Cadima, J. (2016). "Principal component analysis: a review and recent developments." Philosophical Transactions of the Royal Society A, 374, 20150202.); PCA is not EFA - components are composites, not reflective factors (Frick, S., Killisch, R., BiSSantz, N., Wetzel, E. (2025). "PCA Is Not EFA." European Journal of Psychological Assessment, 41(6), 425-426.); Kaiser eigenvalue $>$ 1 rule tends to over-extract (Zwick, W. R., Velicer, W. F. (1986). "Comparison of five rules for determining the number of components to retain." Psychological Bulletin, 99(3), 432-442.).